

MIYAKONOJO, Japan

WMO: 478290

Lat: 31.7300N

Lon: 131.0817E

Elev: 511

StdP: 14.43

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	27.8	29.8	16.3	12.8	41.1	19.0	14.6	40.2	14.0	47.3	12.5	47.2	2.6	0	0.357

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	14.1	92.0	77.4	90.3	76.8	88.5	76.3	79.3	87.0	78.5	85.9	77.8	84.8	7.0	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.4	145.5	82.0	76.6	141.5	81.5	75.9	138.0	81.2	43.4	87.4	42.5	86.2	41.7	84.8	82.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.0	12.0	10.8	DB	23.4	95.0	2.0	1.5	22.0	96.1	20.8	96.9	19.7	97.8	18.2	98.9	(4)
(5)			WB	21.9	80.7	2.0	1.0	20.5	81.4	19.3	82.0	18.2	82.6	16.8	83.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	63.0	43.4	46.4	52.0	60.4	68.2	73.4	81.0	81.8	76.6	67.4	56.8	47.1	(6)
(7)		DBStd	14.13	5.19	6.38	5.91	5.26	3.78	4.20	3.18	2.38	4.19	5.24	6.38	5.96	(7)
(8)		HDD50	545	216	138	46	2	0	0	0	0	0	0	11	132	(8)
(9)		HDD65	2605	669	522	404	154	13	0	0	0	0	34	254	555	(9)
(10)		CDD50	5280	12	37	108	313	566	702	960	985	799	540	216	42	(10)
(11)		CDD65	1861	0	0	0	14	114	252	495	520	349	108	9	0	(11)
(12)		CDH74	15075	0	0	4	89	619	1451	4774	5119	2498	500	21	0	(12)
(13)		CDH80	4828	0	0	0	4	92	310	1738	1910	714	60	0	0	(13)
(14)	Wind	WSAvg	4.4	3.7	4.1	4.6	4.7	4.6	4.9	5.0	4.9	4.7	4.4	3.5	3.6	(14)
(15)	Precipitation	PrecAvg	103.2	3.0	4.5	6.7	8.1	8.5	22.2	13.4	11.2	12.1	6.0	4.4	3.1	(15)
(16)		PrecMax	162.9	8.6	9.1	14.0	16.4	14.3	42.3	44.1	28.3	29.1	18.0	10.8	7.9	(16)
(17)		PrecMin	65.2	0.3	1.5	2.3	2.0	2.6	8.9	0.8	1.9	0.6	1.4	0.1	0.1	(17)
(18)		PrecStd	20.2	1.7	2.4	2.6	3.6	3.1	8.0	9.0	6.5	7.2	3.8	2.5	2.0	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	63.8	69.0	72.9	80.1	86.0	88.6	94.4	94.2	90.9	84.7	76.5	67.7	(19)
MCWB			55.6	60.1	57.9	63.8	65.7	74.4	77.7	77.9	76.5	72.3	67.3	59.0	(20)	
2%		DB	59.9	64.8	69.5	76.6	82.5	86.0	92.2	92.2	88.4	81.5	73.2	64.0	(21)	
		MCWB	52.5	57.7	57.6	61.2	65.1	73.4	77.3	77.6	75.1	69.7	63.6	56.4	(22)	
5%		DB	57.0	61.7	66.3	73.7	80.0	83.7	90.5	90.6	86.5	79.0	70.4	61.2	(23)	
		MCWB	50.1	54.8	56.6	60.5	64.5	72.3	77.1	77.1	74.2	68.0	62.3	54.2	(24)	
10%		DB	54.2	58.3	63.4	70.9	77.5	81.2	88.5	88.8	84.4	76.7	67.9	58.3	(25)	
		MCWB	47.5	51.0	55.0	60.3	64.3	71.5	76.7	76.7	73.5	66.8	60.8	52.1	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.5	63.0	64.9	70.2	73.8	77.9	80.7	80.7	79.1	75.6	70.0	62.7	(27)
MCD B			62.1	66.2	68.3	73.7	78.2	83.5	89.0	88.5	86.7	80.6	74.1	65.0	(28)	
2%		WB	55.3	59.9	61.6	67.6	72.1	76.7	79.2	79.5	77.7	73.7	66.9	59.1	(29)	
		MCD B	57.7	63.4	65.4	70.7	75.7	82.1	87.1	87.3	84.3	78.5	70.2	62.2	(30)	
5%		WB	51.7	56.5	59.3	65.0	70.1	75.6	78.4	78.7	76.6	71.8	64.7	55.9	(31)	
		MCD B	55.3	60.4	63.8	69.7	73.9	80.4	85.8	86.4	82.4	76.4	68.3	59.8	(32)	
10%		WB	48.4	52.6	56.7	62.7	68.2	74.6	77.7	78.0	75.6	69.5	62.3	53.1	(33)	
		MCD B	52.8	56.4	61.7	68.1	73.2	78.9	84.7	85.2	80.9	74.0	66.5	56.9	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	18.6	18.7	18.9	19.0	17.5	12.6	13.5	14.1	14.7	16.8	18.3	18.7	(35)
MCDBR			22.2	21.5	22.5	23.6	22.7	16.8	17.0	17.0	16.5	18.4	19.5	21.2	(36)	
MCWBR			14.9	14.1	12.5	11.3	9.4	6.5	5.7	5.7	6.0	8.2	10.9	13.8	(37)	
5% WB		MCD BR	17.8	17.0	17.3	15.1	13.7	11.1	14.3	14.5	13.3	13.1	13.9	16.5	(38)	
		MCWBR	14.6	13.3	12.6	10.1	8.6	5.7	5.6	5.4	5.7	7.0	9.4	12.5	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.418	0.462	0.524	0.545	0.555	0.557	0.539	0.528	0.513	0.478	0.430	0.406	(40)	
taud		2.111	2.018	1.867	1.865	1.906	1.983	2.067	2.089	2.091	2.107	2.168	2.162	(41)		
Ebn at Noon		246	247	241	243	241	239	243	244	242	240	241	243	(42)		
Edh at Noon		41	49	62	64	62	57	53	51	49	45	39	37	(43)		
(44)	All-Sky Solar Radiation	RadAvg	821	1002	1307	1508	1556	1257	1668	1677	1376	1143	878	764	(44)	
(45)		RadStd	68	89	89	154	174	206	192	120	151	131	87	66	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (20 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page